

Bridged PPMI Project 196 and 222 Olink® Explore 1536 targeted proteomics assessment of CSF and blood plasma

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PPMI Project ID: 9000

Summary

PPMI targeted proteomics projects #196 and #222 feature two separate Targeted Proteomics datasets. These two datasets were generated using Olink 1536 from two separate data providers. For Project #196, analysis was conducted using Olink Explore for 74 participants run on 222 CSF samples & 222 matched plasma samples. For Project #222, analysis was conducted using Olink Explore on 666 CSF samples & 898 plasma samples derived from 225 participants. Both projects included bridging samples intended to be used to normalize combined results upon the application of a bridging protocol. For additional information on project #196 or project #222, including the data dictionary, see their respective methods documents in LONI.

Project 196 & 222 Targeted Proteomics datasets each contain eight unfiltered NPX files from four separate Panels for both Plasma and CSF samples. The four targeted proteomics panels are Cardiometabolic, Inflammation, Neurology, and Oncology.

This project, provided for PPMI users' convenience, applies an Olink-provided bridging protocol, applied in R. For consistency with standard Olink file conventions, the resultant data has been split into 8 component files based on tissue source (CSF, Plasma) and panel (Cardiometabolic, Inflammation, Neurology, and Oncology). Various checks were performed after bridging to ensure the new, bridged data set was as expected.

Use Note: Project 9000 is provided for PPMI users' convenience. Investigators should carefully review the information provided on the bridging method and confirm that the analytical methods applied meet their research objectives. Further, investigators interested in combining Project 9000 with other Olink results should strongly consider reaching out to Olink Biostatistics for guidance.

Method:

R Package *OlinkAnalyze* version 3.6.2 was used in this bridging analysis. The *OlinkAnalyze* collection of functions is developed by Olink and intended to facilitate analysis of proteomics

data from Olink, primarily NPX data. Full details of the Olink Analyze R package can be found [on CRAN](#), including information on code dependencies, limitations of the analysis, and other user guidance. The bridging and mean calculations were all conducted using R and were performed separately for cerebrospinal fluid data sets and the plasma data sets.

Inclusion of Bridging Samples:

The final goal was to perform a joint analysis of datasets from project #196 and project #222, which requires the application of a bridging protocol to remove technical variation between the batches. Olink recommends the inclusion of a minimum of 16-24 of these “bridging samples” across batches, whereupon the protein levels measured for these samples can be used as a common reference between batches in order to normalize and alleviate any potential bias. In this case, 89 samples were included as bridging samples between Projects 196 and 222 for both CSF (Figure 1a) and Plasma (Figure 1b).

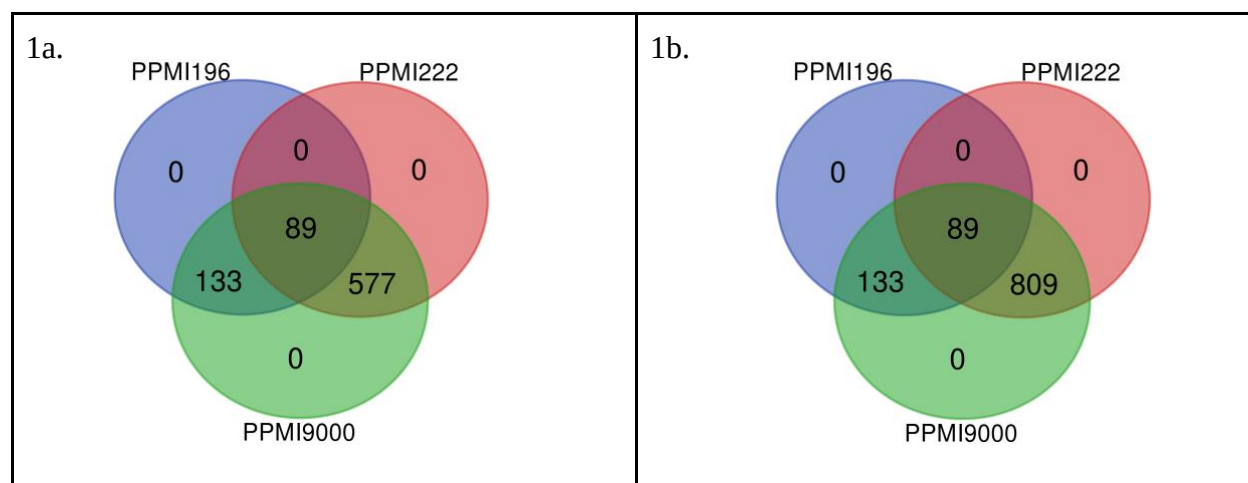


Figure 1. Looking at unique PPMI ID x Visit for shows the 89 common samples that were used for bridging and that every PPMI patient sample from Project 196 and Project 222 is included in Project 9000 for 1a) CSF and 1b) Plasma.

For the purpose of applying the bridging protocol, these samples were identified using the dplyr R package to find the intersecting samples between Project 196 and 222. Upon manual review of results, a formatting difference was identified in 2 samples’ identifying information; once confirmed as a formatting error, this difference was manually corrected by the analyst to ensure the bridging samples were correctly processed by the bridging protocol. This list of bridging samples was used as an input to the `olink_normalization`.

Bridging Data and QC

As described in the *OlinkAnalyze* user manual, the bridging protocol applied involved the calculation of the median of the paired NPX differences per assay using the bridging samples. This determines the assay specific adjustment factor which is used to normalize NPX values

between the two datasets. For Project 9000, Project 196 was considered the reference dataset and therefore its NPX values remained unaltered. Project 222 was adjusted to the reference dataset based on the adjustment factors. The bridging samples themselves are adjusted to their average values and de-duplicated, resulting in one entry per bridging sample in the final bridged dataset. Therefore, users should expect that Project 9000 contains *fewer* records than the combined unbridged results of the component projects.

After applying the bridging protocol, the combined file was reviewed to confirm that the resultant changes to NPX values were consistent with expectations. Based on the assignment of reference (#196) and adjustment (#222) datasets, it was expected that (1) all adjustment dataset NPX values would see change, (2) all bridging samples common between #196 and #222 would see change, and (3) the total record count would decrease when compared to strictly combining #196 and #222 without the bridging protocol. As shown in Table 1 and Figure 2 below, the bridged file meets expectations in this regard when NPX changes are reviewed in total.

List names (CSF)	number of unique elements (PPMIID_Visit_Uniprot_NPX)
PPMI196	88312
PPMI222	251707
PPMI9000	291633

Table 1. Unique NPX associated elements in each project set for the CSF Inflammation data set.

List names (Plasma)	number of unique elements (PPMIID_Visit_Uniprot_NPX)
PPMI196	88312
PPMI222	338550
PPMI9000	376309

Table 2. Unique NPX associated elements in each project set for the Plasma Inflammation data set.

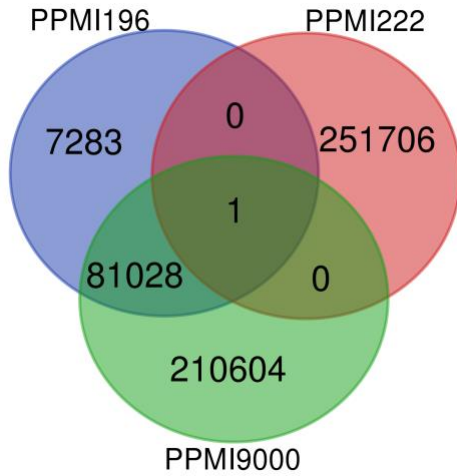


Figure 2: Looking at unique values (combined PPMI ID, visit, Uniprot ID and NPX value into one unique value), of the 88,312 values in project #196, 7283 values were associated with bridging sample values that had changed with normalization. 81,028 values from project #196 were retained after bridging in project #9000 as expected, since project #196 was the reference data set. As expected, 251,706 of the 251,707 unique values from project #222 did not match any of the new NPX values, since project #222 NPX values were changed by a specific normalization factor calculated by the bridging function.

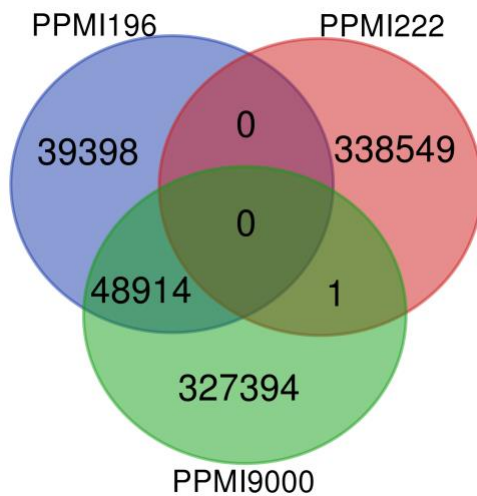


Figure 3: Looking at unique values (combined PPMI ID, visit, Uniprot ID and NPX value into one unique value), of the 88,312 values in project #196, 39398 values were associated with bridging sample values that had changed with normalization. 48,914 values from project #196 were retained after bridging in project #9000 as expected, since project #196 was the reference data set. As expected, 338,549 of the 338,550 unique values from project #222 did not match any of the new NPX values, since project #222 NPX values were changed by

a specific normalization factor calculated by the bridging function.

In some cases, the application of the bridging protocol led to the duplication of the same sample x uniprot with two separate NPX values. In addition, there are cases in which there is a duplicate uniprot ID due to the same uniprot ID being in more than one Olink panel or there being two different Olink assays that map to that uniprot ID. In all of these cases, after the bridging protocol had been applied, the mean NPX value was calculated and used in the final data product, resulting in one NPX value for every uniprot ID for every sample ID.

Analysis of the bridged file using principal components analysis (PCA) was evaluated and found to be consistent with expectations, as seen below.

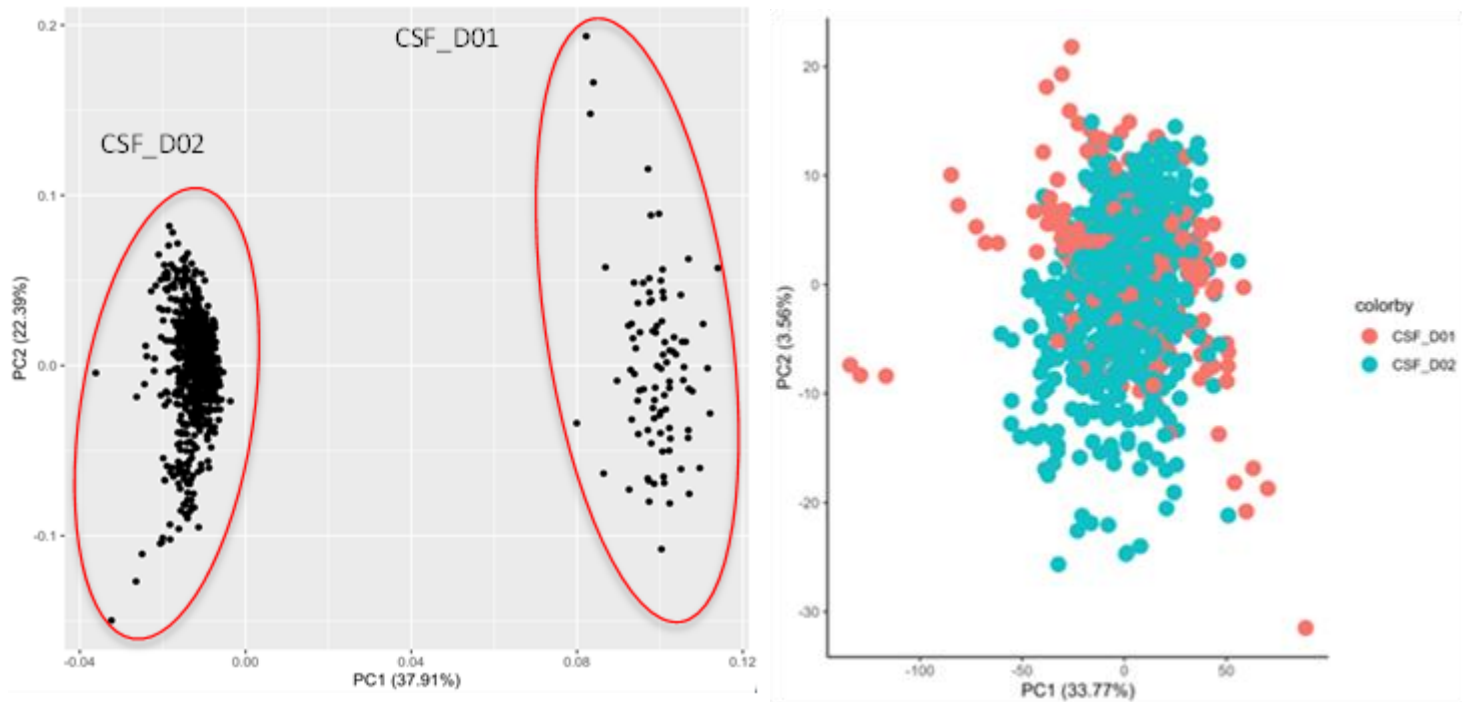


Figure 3. Results of the bridging can be seen through PCA. Figure 3a shows dataset D01 (Project 196 CSF) and D02 (Project 222 CSF) prior to bridging. Separation along PC1 shows datasets D01 and D02 as distinct, with 37.91% of the variation due to batch effects. Figure 3b shows that after performing bridging, D01 and D02 no longer show separation along PC1, indicating that bridging was successful and batch effects were removed.

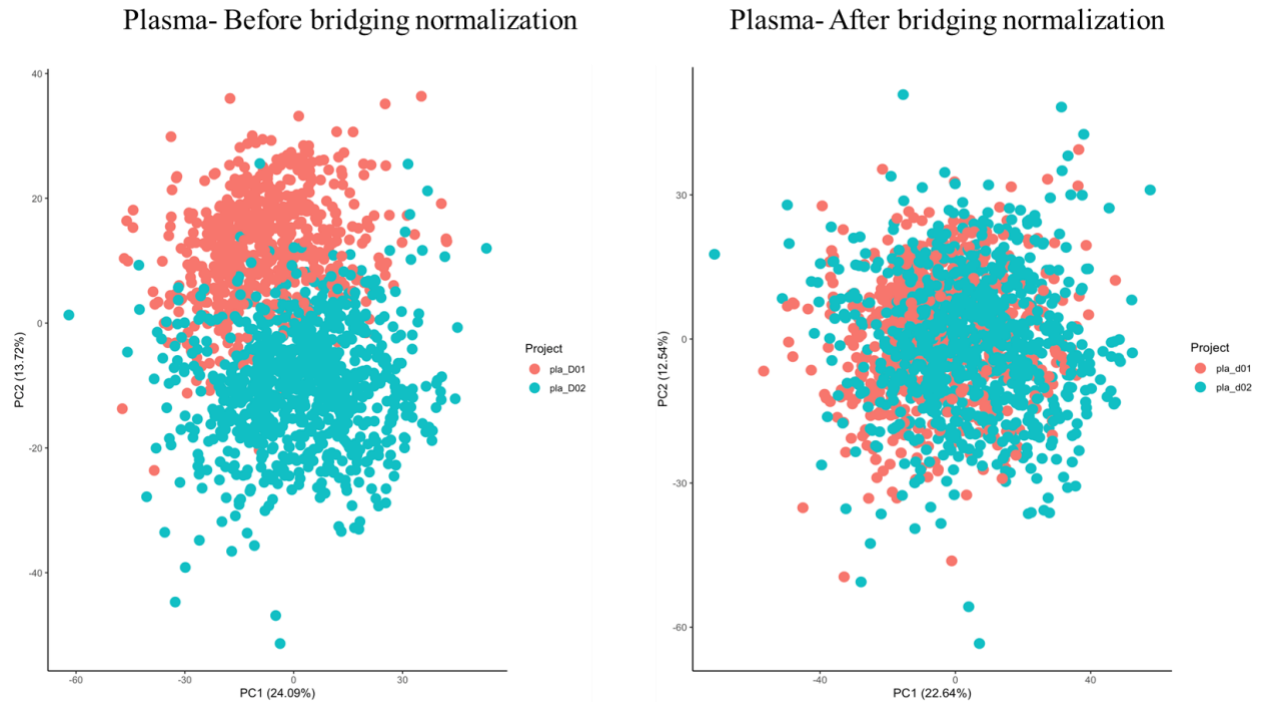


Figure 4. PCA of plasma data before and after bridging normalization shows removal of batch effects by use of bridging normalization.

Appendix I: Analysis Code

Note variables in here can be changed for your specific csv files.

```
##load libraries
```

```
library(pacman)
```

```
p_load("OlinkAnalyze", "dplyr", "data.table", "tidyr", "ggplot2", "stringr", "data.table", "reshape2", "gplots",
"readxl", "magrittr", "tidyverse")
```

```
##CSF bridging normalization
```

```
##read in file 1
```

```
csf_d01 <- read_NPX(filename = "project196.csv")
```

```
##read in file 2
```

```
csf_d02 <- read_NPX(filename = "project222.csv")
```

```
##find overlapping sample IDs in CSF
```

```
##NOTE: change SampleID column to whatever your unique identifier column is
```

```
csf_overlap_samples <- dplyr::intersect(unique(csf_d01$SampleID), unique(csf_d02$SampleID))
```

[illegible]

```

# write.csv(bridged_csf_df, file = "~/Desktop/Bridged_CSF.csv", row.names = F, quote = F)

##Plasma bridging

##read in file 1

# pla_d01 <- read_NPX(filename = "Plasma_D01.csv")

##read in file 2

# pla_d02 <- read_NPX(filename = "Plasma_D02.csv")

##find overlapping sample IDs in Plasma

##NOTE: change SampleID column to whatever your unique identifier column is

pla_overlap_samples <- dplyr::intersect(unique(pla_d01$SampleID), unique(pla_d02$SampleID))

pla_overlap_samples_list <- list("DF1" = pla_overlap_samples,
                                "DF2" = pla_overlap_samples)

##perform bridging function for the two datasets

bridged_pla <- olink_normalization(project_1_df = pla_d01,
                                   project_2_df = pla_d02,
                                   bridge_samples = pla_overlap_samples_list,
                                   project_1_name = "PLA_D01",
                                   project_2_name = "PLA_D02",
                                   project_ref_name = "PLA_D01")

bridged_pla$IDAssay <- paste0(bridged_pla$SampleID, ":", bridged_pla$Assay)

bridged_pla$IDProject <- paste0(bridged_pla$SampleID, ":", bridged_pla$Project)

bridged_pla$IDAssayProject <- paste0(bridged_pla$SampleID, ":", bridged_pla$Assay, ":",
bridged_pla$Project)

##find any duplicated IDs or NPX values, move them to a new table and perform mean calculation

table(duplicated(bridged_pla$IDAssayProject))

bridged_pla_avg <- bridged_pla %>%

  group_by(IDAssay) %>%

  mutate(Avg_NPX = mean(NPX)) %>%

  distinct(IDAssay, .keep_all = TRUE)

```



```
##add in new averaged values into main table
```

```
bridged_pla_avg$UID <- paste0(bridged_pla_avg$participant_id, "_", bridged_pla_avg$visit_month, "_",  
bridged_pla_avg$Project)
```

```
## write csv file of final data
```

```
write.csv(bridged_pla_avg, file = "~/Desktop/Bridged_PLA_Olink_format.csv", row.names = F, quote = F)
```

```
##create data matrix and write csv data matrix
```

```
bridged_pla_df <- reshape2::dcast(data = bridged_pla_avg,
```

```
    formula = Assay~UID,
```

```
    value.var = "Avg_NPX", drop = F)
```

```
write.csv(bridged_pla_df, file = "~/Desktop/Bridged_PLA.csv", row.names = F, quote = F)
```